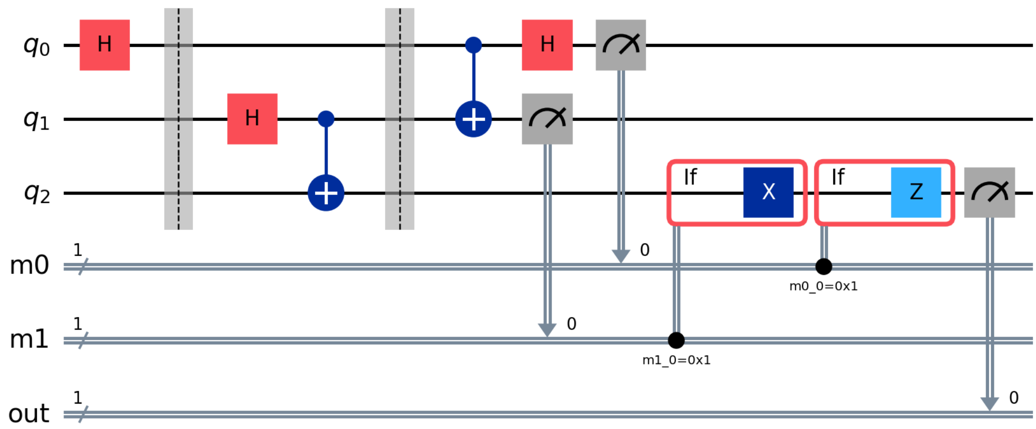
SECOND LAB EXERCISE

Entanglement, superdense coding and quantum teleportation



Python implementation with Qiskit

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Academic year	2025-2026

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1. Objective and methodology

The second practice was implemented with Qiskit Aer. Every measured experiment uses 1000 shots. SciPy performs the chi-square tests for Bell-state histograms and the two-sided binomial tests for balanced single-qubit measurements. A fixed random seed makes the report reproducible.

2. Bell states and statistical validation

The four Bell states have two possible computational-basis outcomes with theoretical probability 0.5 each. The chi-square test compares the observed pair of counts with the expected [500,500]. All p-values are greater than 0.05, so the fluctuations are compatible with the theoretical distributions.

State	Counts	Chi-square	p-value
Phi+	{'00': 504, '01': 0, '10': 0, '11': 496}	0.064	0.8003
Phi-	{'00': 510, '01': 0, '10': 0, '11': 490}	0.400	0.5271
Psi+	{'00': 0, '01': 500, '10': 500, '11': 0}	0.000	1.0000
Psi-	{'00': 0, '01': 524, '10': 476, '11': 0}	2.304	0.1290

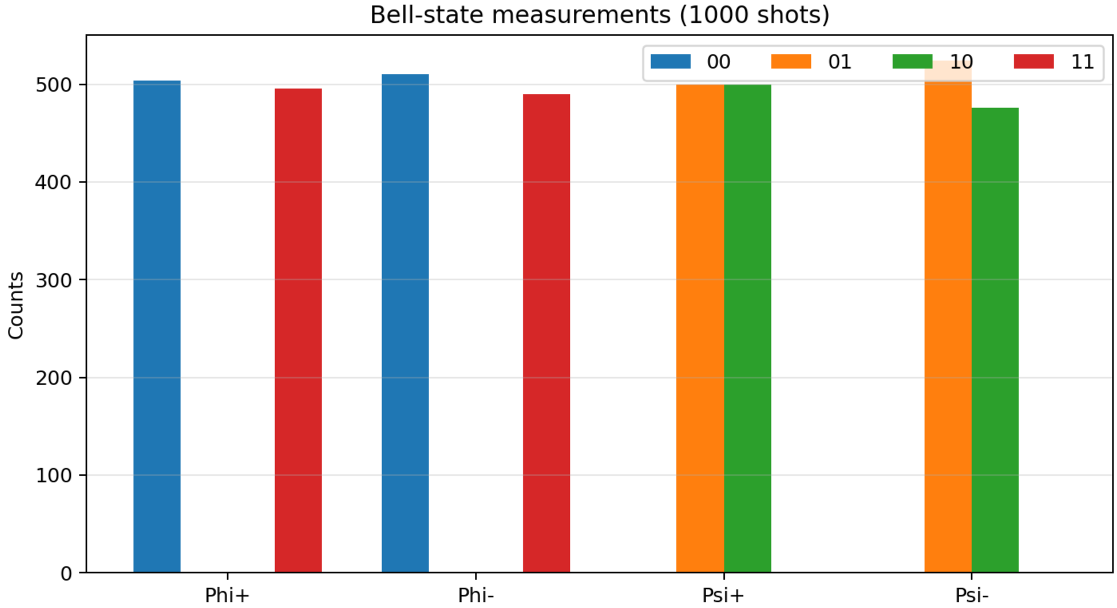

Figure 1. Simulated measurements of the four Bell states.

2.1 Circuit construction

```
        qc.x(1)
    if measured:
        qc.measure([0, 1], [0, 1])
    return qc


def superdense_circuit(message: str) -> QuantumCircuit:
    """Encode message b1b0 with Z^b1 X^b0 on Alice's qubit."""
    qc = QuantumCircuit(2, 2, name=f"SDC {message}")
    qc.h(0)
    qc.cx(0, 1)
    if message[1] == "1":
        qc.x(0)
    if message[0] == "1":
        qc.z(0)
    qc.cx(0, 1)
    qc.h(0)
    qc.measure([0, 1], [0, 1])
```

```
    return qc


def prepare_state(qc: QuantumCircuit, qubit: int, name: str) -> None:
    if name == "0":
        return
    if name == "1":
        qc.x(qubit)
    elif name == "+":
```

3. Superdense coding

Alice and Bob first share Phi+. Alice encodes two classical bits by applying Z^b1 X^b0 to her qubit. The receiver applies CNOT and H, converting the Bell basis back into a computational-basis message. In every case the most frequent measured bit string equals the transmitted message.

Message	Decoded	Counts
00	00	{'00': 1000}
01	01	{'01': 1000}
10	10	{'10': 1000}
11	11	{'11': 1000}

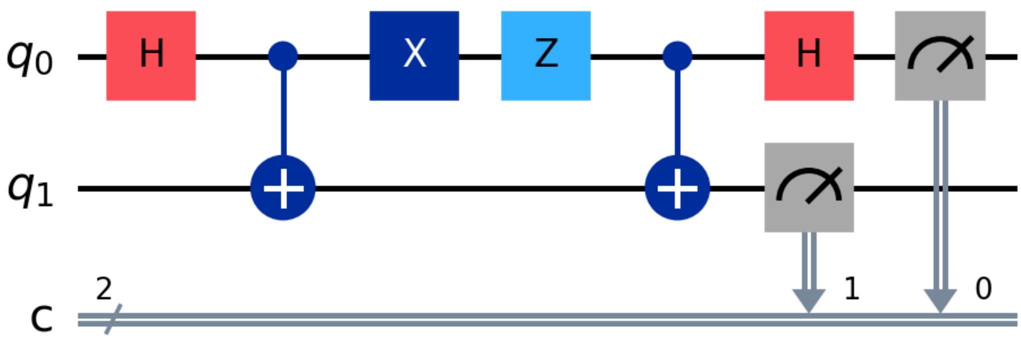

Figure 2. Superdense-coding circuit for message 11.

4. Quantum teleportation

The circuit creates an entangled pair between Alice and Bob. Alice performs a Bell-basis transformation and measures two qubits. The classical outcomes control X and Z corrections on Bob's qubit. Qiskit control-flow blocks implement the classical feed-forward explicitly.

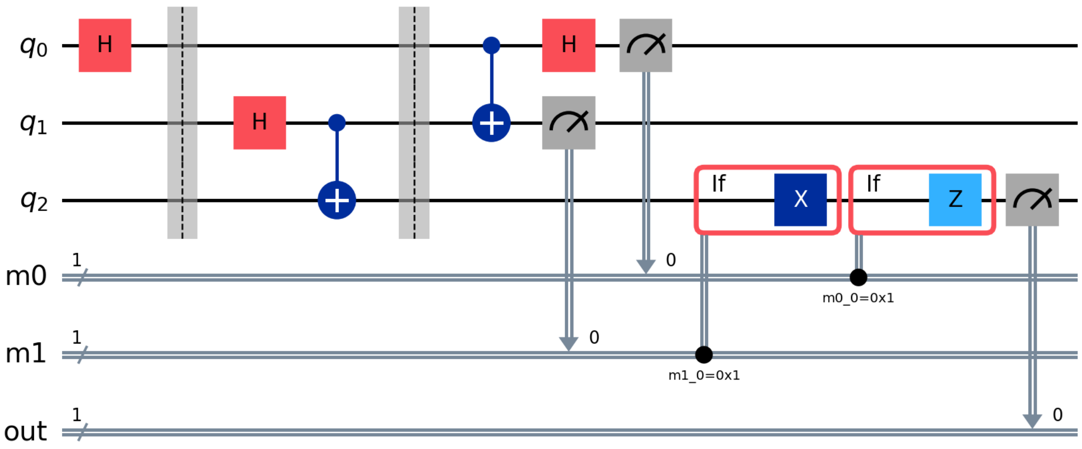

Figure 3. Dynamic teleportation circuit with classical corrections.

Input	Counts	Expected P(1)	p-value if 50/50	Fidelity
|0>	{'0': 1000, '1': 0}	0.00	-	1.000000
|1>	{'0': 0, '1': 1000}	1.00	-	1.000000
|+>	{'0': 492, '1': 508}	0.50	0.6353	1.000000
|->	{'0': 493, '1': 507}	0.50	0.6810	1.000000
|i>	{'0': 494, '1': 506}	0.50	0.7280	1.000000
|-i>	{'0': 495, '1': 505}	0.50	0.7760	1.000000

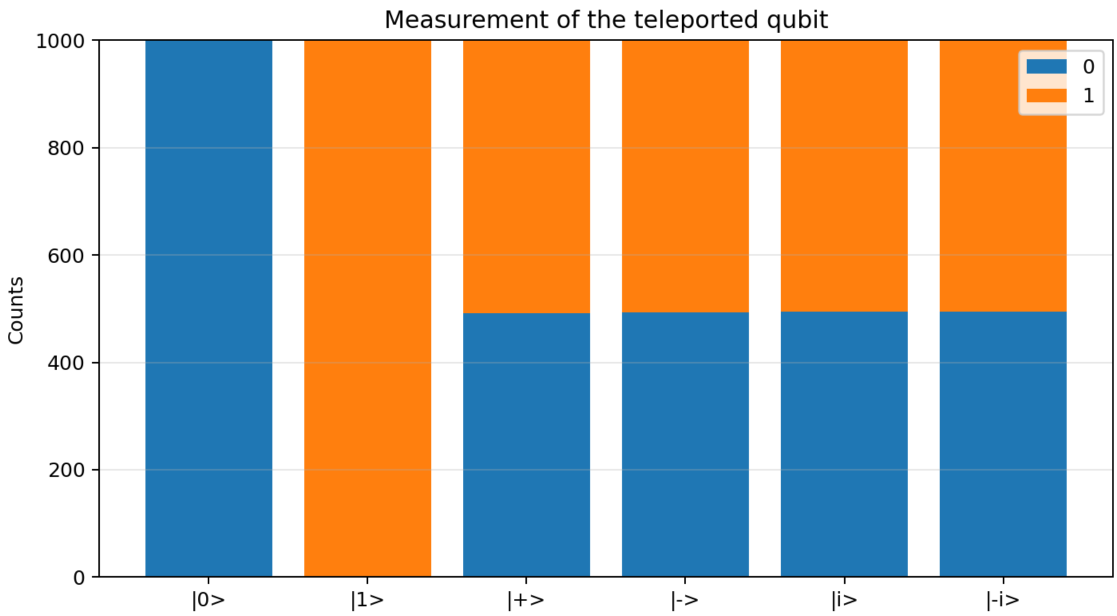

Figure 4. Computational-basis measurements of the teleported states.

The states |0> and |1> are measured deterministically. The four equatorial states produce balanced counts. Their binomial-test p-values do not indicate a statistically significant departure from 0.5. The deferred-measurement verification gives unit fidelity for all six input states.

5. Additional Hadamard test

Applying H after teleportation changes the measurement basis. H|+>=|0> and H|->=|1>, so these two states become deterministic. H|0> and H|1> become balanced states. The |i> and |-i> states also remain balanced in this basis.

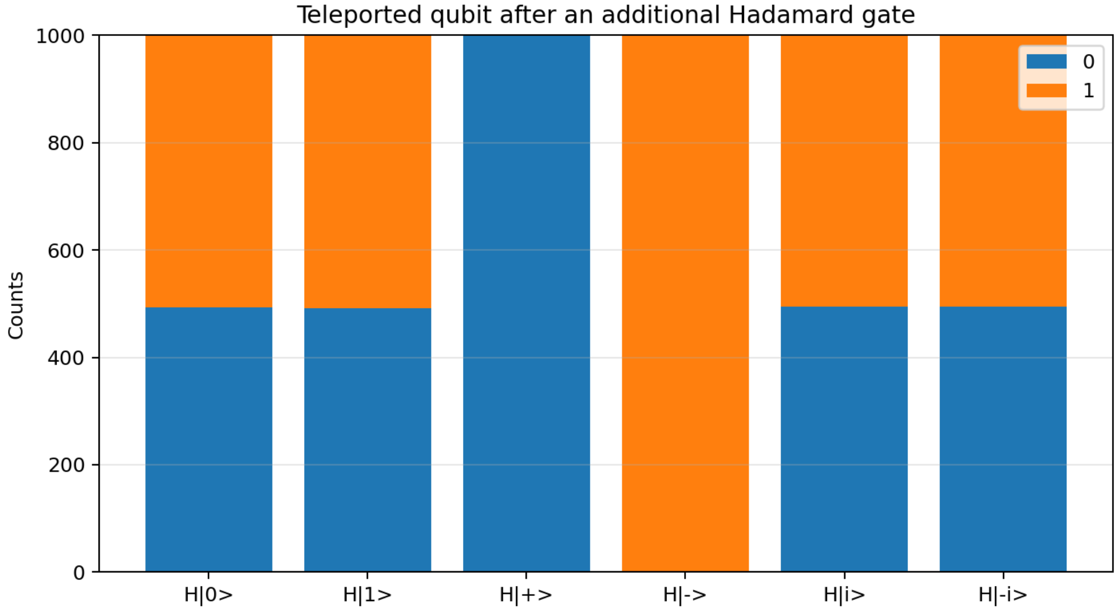

Figure 5. Measurements after applying an additional Hadamard gate.

6. Conclusions

The simulations verify entanglement, superdense coding and teleportation. The statistical tests distinguish ordinary finite-shot fluctuations from a meaningful disagreement with theory. Dynamic classical control does not copy the unknown state; instead, entanglement and two classical bits transfer the state to Bob while Alice's original state is destroyed by measurement.

7. Files and execution

Run `python iqc2_entanglement_teleportation.py`. Results are stored in `IQC2_results.json`. The Jupyter notebook runs the same implementation interactively.